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ADYAPAN

Adyapan School

BioPython

Duration - 2 months

Industry Certification	Skill India कौशल भारत-कुशल भारत · N·S·D·C National Skill Development Corporation · Transforming the skill landscape **Skill India Certified**	**250+ Partner Companies**

Adyapan School

From code to biological discovery master Biopython.

From biological data to real-world insights - become industry-ready.

This immersive 8-week Biopython program takes you beyond foundational concepts into practical bioinformatics workflows. Learn sequence analysis, database integration, alignment techniques, and genomic data processing through hands-on projects and guided pipelines. With a strong focus on real-world applications and computational biology practices, you'll graduate with the ability to analyze biological data, build automated bioinformatics pipelines, and derive meaningful insights for research and industry use.

8	30+	20,000+
MODULES	PROGRAM OFFERINGS	STUDENTS

250+ PARTNERED COMPANIES

ABOUT ADYAPAN SCHOOLS

Where education meets real-world impact

Not just a course — a platform to launch your career.

Adyapan Schools was built with a single conviction: learning works best when it happens in the real world. We partner with top companies, mentors, and industry platforms to ensure every student graduates with a portfolio of work that speaks louder than a certificate.

Our programs combine rigorous coursework with live client projects, giving you the skills and proof-of-work that employers actually want.

MISSION

To equip ambitious learners with practitioner-level digital marketing skills through mentor-led, project-based education that bridges the gap between learning and earning.

VISION

To be India's most trusted launchpad for the next generation of marketing leaders — defined not by degrees but by the real work.

Adyapan School

Everything you need to grow fast

PROGRAM HIGHLIGHTS

Live Industry Projects

Work on campaigns for real brands alongside your coursework. Build portfolio projects that prove your expertise to employers.

1-on-1 Mentorship

Dedicated mentors from Google, Microsoft, Mastercard and more. Get personalized guidance and industry connections.

AI-Powered Marketing

Learn cutting-edge AI tools alongside evergreen fundamentals. Stay ahead of the curve in a rapidly evolving landscape.

Dual Certification

Earn both a Course Completion and Internship Certificate — accredited by Skill India Digital Hub and NSDC.

Internship Guarantee

Graduate with an internship completion certificate from a live brand project. Concrete, resume-ready proof of work.

Industry Network

Join a network of alumni at Amazon, Google, Adobe, Microsoft. Access exclusive hiring events and referral opportunities.

CURRICULUM

8 weeks. 8 modules. Infinite impact.

WEEK 1

Python Foundations & Biopython Setup

- Discussion of curriculum
- Review core Python concepts
- Install and configure Biopython using pip and Conda; verify explore the module structure
- Understand the Biopython ecosystem
- Work with the Seq object
- Perform basic sequence operations
- Understand MutableSeq vs Seq objects and when to use each in practical workflows

WEEK 2

Sequence Parsing & File Format Handling

- Understand the most common biological file formats
- Use Bio.SeqIO to parse single and multi-record files; iterate over SeqRecord objects efficiently
- Extract metadata from SeqRecord objects
- Convert between file formats using SeqIO.convert() and write processed records to new output files
- Handle large sequence files using SeqIO.index() and SeqIO.index_db() for memory-efficient access

WEEK 3

Accessing Biological Databases with Entrez & ExPASy

- Understand the NCBI Entrez system and the databases it provides: PubMed, Nucleotide, Protein, Gene, and SRA
- Use Bio.Entrez to programmatically search, fetch, and parse records from NCBI using esearch and efetch
- Download GenBank and FASTA records for genes or proteins of interest and parse them using SeqIO
- Query the KEGG database using Bio.KEGG
- Build automated pipelines to batch-download and organise sequences from NCBI for downstream analysis

CURRICULUM

8 weeks. 8 modules. Infinite impact.

WEEK 4

Sequence Alignment & BLAST

- Understand pairwise sequence alignment theory
- Perform pairwise alignments and interpret alignment scores and gap penalties
- Use Bio.pairwise2 for legacy pairwise alignment workflows and compare outputs between methods
- Run BLAST searches programmatically
- Extract hit descriptions, e-values, bit scores, and aligned sequences from BLAST result objects
- Perform local BLAST searches using subprocess calls to a locally installed BLAST+ binary for large datasets

WEEK 5

Multiple Sequence Alignment & Phylogenetics

- Multiple sequence alignment (MSA)
- Run ClustalW, MUSCLE, and MAFFT through Biopython wrappers using Bio.Align.Applications
- Parse and manipulate MSA outputs using Bio.AlignIO
- Build phylogenetic trees using distance-based methods with Bio.Phylo and TreeConstruction
- Visualise and annotate phylogenetic trees using Bio.Phylo.draw() and export in Newick and Nexus formats

WEEK 6

Working with Genomic Features & Annotation

- Parse GenBank and GFF3 files to extract annotated features
- Use SeqFeature objects to locate features by type, qualifier, and genomic coordinates
- Understand FeatureLocation and CompoundLocation for handling multi-exon and strand-aware features
- Perform coordinate conversions and map features
- Cross-reference genomic features with external databases (using programmatic lookups

CURRICULUM

8 weeks. 8 modules. Infinite impact.

WEEK 7

Structural Bioinformatics & Population Genetics

- Introduction to the PDB file format and the biological information encoded in ATOM and HETATM records
- Parse PDB structures using Bio
- Calculate structural properties and B-factor profiles
- Perform structural superimposition using the Superimposer module and calculate RMSD between structures
- Introduction to population genetics with Biopython

WEEK 8

Capstone Project & Real-World Pipelines

- Integrate all concepts and design and implement a complete end-to-end bioinformatics pipeline using Biopython from scratch
- Apply best practices in Python scripting: modular code, error handling, logging, and argparse-based CLI tools
- Integrate multiple Biopython modules in a single automated workflow with well-documented functions

Adyapan School

WHO THIS IS FOR

This course is perfect for

- Students & Career Switchers
- Biotechnology, Microbiology & Life Science Students
- Researchers & Lab Professionals
- Python Programmers Entering Bioinformatics
- Healthcare, Genomics & Drug Discovery Enthusiasts
- Aspiring Bioinformatics Professionals

CERTIFICATIONS

ALUMNI NETWORK

Our alumni work at world-class companies

Amazon | Adobe | Google | Autodesk | Microsoft | Deloitte

Your career switch is one click away.

Ready to begin? Apply at **adyapanschool.com** or email us at **support@adyapan.com**

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